main.r

Run ▶

```r
# Q1: Approximate Median

age <- c("1-5", "5-15", "15-20",
         "20-50", "50-80", "80-110")

freq <- c(200, 450, 300, 1500, 700, 44)

lower <- c(1, 5, 15, 20, 50, 80)
upper <- c(5, 15, 20, 50, 80, 110)

N <- sum(freq)
cf <- cumsum(freq)

median_class <- which(cf >= N / 2)[1]

L <- lower[median_class]
f <- freq[median_class]
cf_prev <- ifelse(median_class == 1, 0,
                  cf[median_class - 1])
h <- upper[median_class] - lower[median_class]

median_value <- L + ((N / 2 - cf_prev) / f) * h

print(N)
```

Output

```
[1] 3194
[1] 32.94
```

main.r

Run ▶

```r

age <- c(13, 15, 16, 16, 19, 20, 20, 21,
         22, 22, 25, 25, 25, 25, 30, 33,
         33, 35, 35, 35, 35, 36, 40, 45,
         46, 52, 70)

# Mean
print(mean(age))

# Median
print(median(age))

# Mode
mode_value <- as.numeric(
  names(which(table(age) == max(table(age))))
)
print(mode_value)

# Modality
print("Unimodal: 25 and 35 are tied as the modes")

```

Output

```
[1] 29.96296
[1] 25
[1] 25 35
[1] "Unimodal: 25 and 35 are tied as the modes"
[1] 41.5
 25%  75%
20.5 35.0
```

main.r

Run ▶

```r
# Q3: Normalization

data <- c(200, 300, 400, 600, 1000)

# (a) Min-Max normalization: range [0, 1]
min_max <- (data - min(data)) /
           (max(data) - min(data))
```

main.r

Run ▶

```r
# Q3: Normalization

data <- c(200, 300, 400, 600, 1000)

# (a) Min-Max normalization: range [0, 1]
min_max <- (data - min(data)) /
           (max(data) - min(data))

print("Min-Max Normalization:")
print(min_max)

# (b) Z-score normalization
z_score <- (data - mean(data)) / sd(data)

print("Z-Score Normalization:")
print(z_score)
```

Output

```
[1] "Min-Max Normalization:"
[1] 0.000 0.125 0.250 0.500 1.000
[1] "Z-Score Normalization:"
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
```

main.r

Run ▶

```r
          40, 45, 45, 45, 71, 72, 73, 75)

# Divide into bins of 3 values
bins <- split(data, ceiling(seq_along(data) / 3))

# (a) Smoothing by bin mean
bin_mean <- lapply(bins, function(x) {
  rep(mean(x), length(x))
})
print("Smoothing by Bin Mean:")
print(unlist(bin_mean))

# (b) Smoothing by bin median
bin_median <- lapply(bins, function(x) {
  rep(median(x), length(x))
})
print("Smoothing by Bin Median:")
print(unlist(bin_median))

# (c) Smoothing by bin boundaries
bin_boundaries <- lapply(bins, function(x) {
  low <- min(x)
  high <- max(x)
  ifelse(abs(x - low) <= abs(x - high), low, high)
})
print("Smoothing by Bin Boundaries:")
print(unlist(bin_boundaries))
```

Output

```
[1] "Smoothing by Bin Mean:"
      11       12       13       21       22       23       31
  32
12.33333 12.33333 12.33333 15.33333 15.33333 15.33333 19.66667
19.66667
      33       41       42       43       51       52       53
  61
19.66667 20.66667 20.66667 20.66667 23.00000 23.00000 23.00000
38.33333
      62       63       71       72       73       81       82
  83
38.33333 38.33333 53.66667 53.66667 53.66667 73.33333 73.333
73.33333
[1] "Smoothing by Bin Median:"
11 12 13 21 22 23 31 32 33 41 42 43 51 52 53 61 62 63 71 72
82 83
13 13 13 15 15 15 20 20 20 21 21 21 23 23 23 40 40 40 45 45
73 73
[1] "Smoothing by Bin Boundaries:"
11 12 13 21 22 23 31 32 33 41 42 43 51 52 53 61 62 63 71 72
82 83
11 13 13 15 15 16 19 20 20 20 21 21 22 22 24 30 45 45 45 45
72 75
```

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main.r

Run ▶

```r
print(mean(fat))
print(median(fat))
print(sd(fat))

# Boxplots
par(mfrow = c(1, 2))
boxplot(age, main = "Boxplot of Age", ylab = "Age")
boxplot(fat, main = "Boxplot of Body Fat", ylab = "% Fat")

# Scatter plot
plot(age, fat,
     main = "Age vs Body Fat",
     xlab = "Age",
     ylab = "Body Fat (%)",
     pch = 19)

# Q-Q plots
par(mfrow = c(1, 2))
qqnorm(age, main = "Q-Q Plot of Age")
qqline(age)

qqnorm(fat, main = "Q-Q Plot of Body Fat")
qqline(fat)

par(mfrow = c(1, 1))
```

Output

```
[1] 34.11111
[1] 33.5
[1] 6.19192
[1] 26.16667
[1] 26
[1] 4.176263
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else
"Rplot%03d.pdf",  :
  cannot open file 'Rplots.pdf'
Calls: par -> <Anonymous>
Execution halted

=== Code Exited With Errors ===
```

main.r

Run ▶

```r
# Q6: Normalization

age_value <- 35

# Replace these with the actual age minimum and maximum
min_age <- 18
max_age <- 60

# (i) Min-Max normalization
min_max <- (age_value - min_age) /
           (max_age - min_age)
print(min_max)

# (ii) Z-score normalization
mean_age <- 35
sd_age <- 12.94

z_score <- (age_value - mean_age) / sd_age
print(z_score)

# (iii) Decimal scaling normalization
decimal_scaling <- age_value / (10^2)
print(decimal_scaling)
```

Output

```
[1] 0.4047619
[1] 0
[1] 0.35
```

main.r

Run ▶

```r
# Q7: Pencil Data

pencils <- c(9, 25, 23, 12, 11,
             6, 7, 8, 9, 10)

# Mean
print(mean(pencils))

# Median
print(median(pencils))

# Mode
mode_value <- as.numeric(
  names(which(table(pencils) == max(table(pencils))))
)
print(mode_value)
```

Output

```
[1] 12
[1] 9.5
[1] 9
```

main.r

Run

```
                    max(sorted_marks),
                    length.out = 4),
       include.lowest = TRUE,
       labels = FALSE)
)

print("Equal-Width Bins:")
print(equal_width_bins)

# Histogram
par(mfrow = c(1, 2))

hist(marks,
     breaks = 3,
     main = "Equal-Width Histogram",
     xlab = "Marks",
     col = "lightblue")
```

Output

```
[1] 50 55 55 55 58 59

$`2`
[1] 60 61 63 63 65

$`3`
[1] 65 67 71 71 72 75

[1] "Equal-Width Bins:"
$`1`
[1] 50 55 55 55 58

$`2`
[1] 59 60 61 63 63 65 65

$`3`
[1] 67 71 71 72 75
```

main.r

Run ▶

```r
# Q10: Speed Data

speed <- c(78.3, 81.8, 82, 74.2, 83.4,
           84.5, 82.9, 77.5, 80.9, 70.6)

# First quartile
Q1 <- quantile(speed, 0.25)

# Third quartile
Q3 <- quantile(speed, 0.75)

# Interquartile range
IQR_value <- IQR(speed)

# Standard deviation
SD_value <- sd(speed)

print(Q1)
print(Q3)
print(IQR_value)
print(SD_value)
```

Output

```
 25%
77.7
   75%
82.675
[1] 4.975
[1] 4.445835
```

main.r

Run ▶

```r

# Q11: Calculate First and Third Quartiles

age <- c(13, 15, 16, 16, 19, 20, 20, 21,
         22, 22, 25, 25, 25, 25, 30, 33,
         33, 35, 35, 35, 35, 36, 40, 45,
         46, 52, 70)

# Calculate first quartile
Q1 <- quantile(age, 0.25)

# Calculate third quartile
Q3 <- quantile(age, 0.75)

# Display results
print("First Quartile (Q1):")
print(Q1)

print("Third Quartile (Q3):")
print(Q3)

```

Output

```
[1] "First Quartile (Q1):"
 25%
20.5
[1] "Third Quartile (Q3):"
75%
 35
```